



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 24/07/2026

EXPERIMENT 10 -- Database with VIEW and INDEX

AIM:

To create **VIEW** and **INDEX** on the student_cgpa table using MySQL Workbench.

PROCEDURE:

1. Open MySQL Workbench and select the required database.
2. Create the student_cgpa table and insert the records.
3. Create a **VIEW** using the required columns.
4. Display the contents of the view.
5. Perform insert and delete operations on the view (if applicable).
6. Create an **INDEX** on the required column.
7. Execute queries using the indexed column.
8. Drop the index after verification.
9. Observe the output and conclude the experiment.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123, 'Deepthi', 'CSE', 22, 2, 8.71),  
(192512121, 'Varun', 'IT', 27, 1, 9.23),  
(192312117, 'Aadhya', 'ECE', 25, 3, 8.97),  
(192411227, 'Dimple', 'CSE', 31, 0, 9.37),  
(192820896, 'Mohan', 'EEE', 18, 9, 6.48),
```

```
( '192411122','Yogitha','CSE',31,1,9.01),
( '192516871','Sadhya','IT',25,5,7.22),
('192111101','Rahul','MECH',30,12,5.14),
('192411010','Alekhy','CSE',35,2,9.01),
('192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:

Result Grid						
reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa	
192111101	Rahul	MECH	30	12	5.14	
192312117	Aadhya	ECE	25	3	8.97	
192411010	Alekhy	CSE	35	2	9.01	
192411122	Yogitha	CSE	31	1	9.01	
192411123	Deepthi	CSE	22	2	8.71	
192411227	Dimple	CSE	31	0	9.37	
192511135	Rahul	CIVIL	37	0	9.72	
192512121	Varun	IT	27	1	9.23	
192516871	Sadhya	IT	25	5	7.22	
192820896	Mohan	EEE	18	9	6.48	

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

VIEW

1. Create a View

```
CREATE VIEW v1 AS
SELECT reg_no, student_name, cgpa
FROM student_cgpa;
```

OUTPUT:

3	12:52:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) V...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
4	12:52:48	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned
5	12:54:58	CREATE VIEW v1 AS SELECT reg_no, student_name, cgpa FROM SELECT * FROM student_cgpa LIMIT 0, 1000	0 row(s) affected

2. Display the View

```
SELECT * FROM v1;
```

OUTPUT:

Result Grid  Filter Rows:			
	reg_no	student_name	cgpa
▶	192111101	Rahul	5.14
	192312117	Aadhya	8.97
	192411010	Alekhya	9.01
	192411122	Yogitha	9.01
	192411123	Deepthi	8.71
	192411227	Dimple	9.37
	192511135	Rahul	9.72
	192512121	Varun	9.23
	192516871	Sadhya	7.22
	192820896	Mohan	6.48

v1 10 ×

INDEX

1. Create an Index on the cgpa column and display students with CGPA greater than 8.50

```
CREATE INDEX index1  
ON student_cgpa(cgpa);
```

```
SELECT student_name, cgpa  
FROM student_cgpa  
WHERE cgpa > 8.50;
```

OUTPUT:

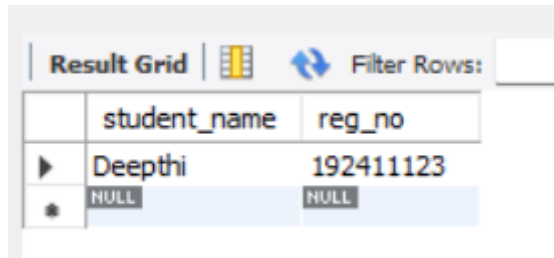
Result Grid  Filter Rows:		
	student_name	cgpa
▶	Deepthi	8.71
	Aadhya	8.97
	Alekhya	9.01
	Yogitha	9.01
	Varun	9.23
	Dimple	9.37
	Rahul	9.72

2. Create an Index on the reg_no column and display the student with Registration Number

```
192411123  
CREATE INDEX index2  
ON student_cgpa(reg_no);
```

```
SELECT student_name, reg_no
FROM student_cgpa
WHERE reg_no = '192411123';
```

OUTPUT:



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains two columns: 'student_name' and 'reg_no'. The first row shows the values 'Deepthi' and '192411123'. The second row shows 'NULL' and 'NULL'.

	student_name	reg_no
▶	Deepthi	192411123
✱	NULL	NULL

RESULT: The records from the tables are displayed using view and index on the given relation.